

Experiment No. 4
Explore the data visualization techniques on seaborn dataset
Date of Performance: 17/02/2025
Date of Submission: 24/02/2025

Aim: Explore Data Visualization techniques on the Titanic Survival dataset.

Objective: Able to apply data visualization techniques on the given dataset.

Theory:

Data visualization is a graphical representation of information and data. By using visual elements like charts, graphs, and maps, data visualization tools provide an accessible way to see and understand trends, outliers, and patterns in data. This blog on data visualization techniques will help you understand detailed techniques and benefits.

In the world of Big Data, data visualization in Python tools and technologies are essential to analyze massive amounts of information and make data-driven decisions.

Benefits of good data visualization

Our eyes are drawn to colours and patterns. We can quickly identify red from blue, and square from the circle. Our culture is visual, including everything from art and advertisements to TV and movies.

Data visualization is another form of visual art that grabs our interest and keeps our eyes on the message. When we see a chart, we quickly see trends and outliers. If we can see something, we internalize it quickly. It's storytelling with a purpose. If you've ever stared at a massive spreadsheet of data and couldn't see a trend, you know how much more effective a visualization can be. The uses of Data Visualization as follows.

- Powerful way to explore data with presentable results.
- Primary use is the pre-processing portion of the data mining process.
- Supports the data cleaning process by finding incorrect and missing values.
- For variable derivation and selection means to determine which variable to include and discard in the analysis.
- Also play a role in combining categories as part of the data reduction process.

Data Visualization Techniques

- Box plots
- Histograms
- Heat maps
- Charts
- Tree maps

- Word Cloud/Network diagram

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from wordcloud import WordCloud


# Load the Titanic dataset

file_path = "/mnt/data/TITANIC.csv"

df = pd.read_csv(file_path)


# Set a visual style

sns.set_style("whitegrid")


# Histograms for numerical features

df.hist(figsize=(12, 8), bins=20, edgecolor='black', layout=(3, 3))

plt.suptitle("Distribution of Numerical Features", fontsize=16)

plt.show()


# Box plots for Age and Fare to identify outliers

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

sns.boxplot(y=df["Age"], ax=axes[0], palette="pastel")

axes[0].set_title("Box Plot of Age")
```

```
sns.boxplot(y=df["Fare"], ax=axes[1], palette="pastel")
```

```
axes[1].set_title("Box Plot of Fare")
```

```
plt.show()
```

```
# Heatmap to show correlations
```

```
plt.figure(figsize=(10, 6))
```

```
sns.heatmap(df.corr(), annot=True, cmap="coolwarm", fmt=".2f", linewidths=0.5)
```

```
plt.title("Correlation Heatmap")
```

```
plt.show()
```

```
# Bar plot for survival rate by Sex
```

```
plt.figure(figsize=(8, 5))
```

```
sns.barplot(x="Sex", y="Survived", data=df, palette="coolwarm")
```

```
plt.title("Survival Rate by Sex")
```

```
plt.ylabel("Survival Rate")
```

```
plt.show()
```

```
# Bar plot for survival rate by Pclass
```

```
plt.figure(figsize=(8, 5))
```

```
sns.barplot(x="Pclass", y="Survived", data=df, palette="coolwarm")
```

```
plt.title("Survival Rate by Passenger Class")
```

```
plt.ylabel("Survival Rate")
```

```
plt.show()
```

```
# Bar plot for survival rate by Embarked location
```

```
plt.figure(figsize=(8, 5))
```

```
sns.barplot(x="Embarked", y="Survived", data=df, palette="coolwarm")
```

```
plt.title("Survival Rate by Embarkation Port")
```

```
plt.ylabel("Survival Rate")
```

```
plt.show()
```

```
# Stacked bar plot for survival by class and gender
```

```
plt.figure(figsize=(8, 5))
```

```
sns.histplot(data=df, x="Pclass", hue="Sex", multiple="stack", palette="coolwarm", bins=3)
```

```
plt.title("Survival Distribution by Class and Gender")
```

```
plt.xlabel("Passenger Class")
```

```
plt.ylabel("Count")
```

```
plt.show()
```

```
# Generate a word cloud for passenger names
```

```
wordcloud = WordCloud(width=800, height=400, background_color="white").generate("".join(df["Name"]))
```

```
# Plot the word cloud
```

```
plt.figure(figsize=(10, 5))
```

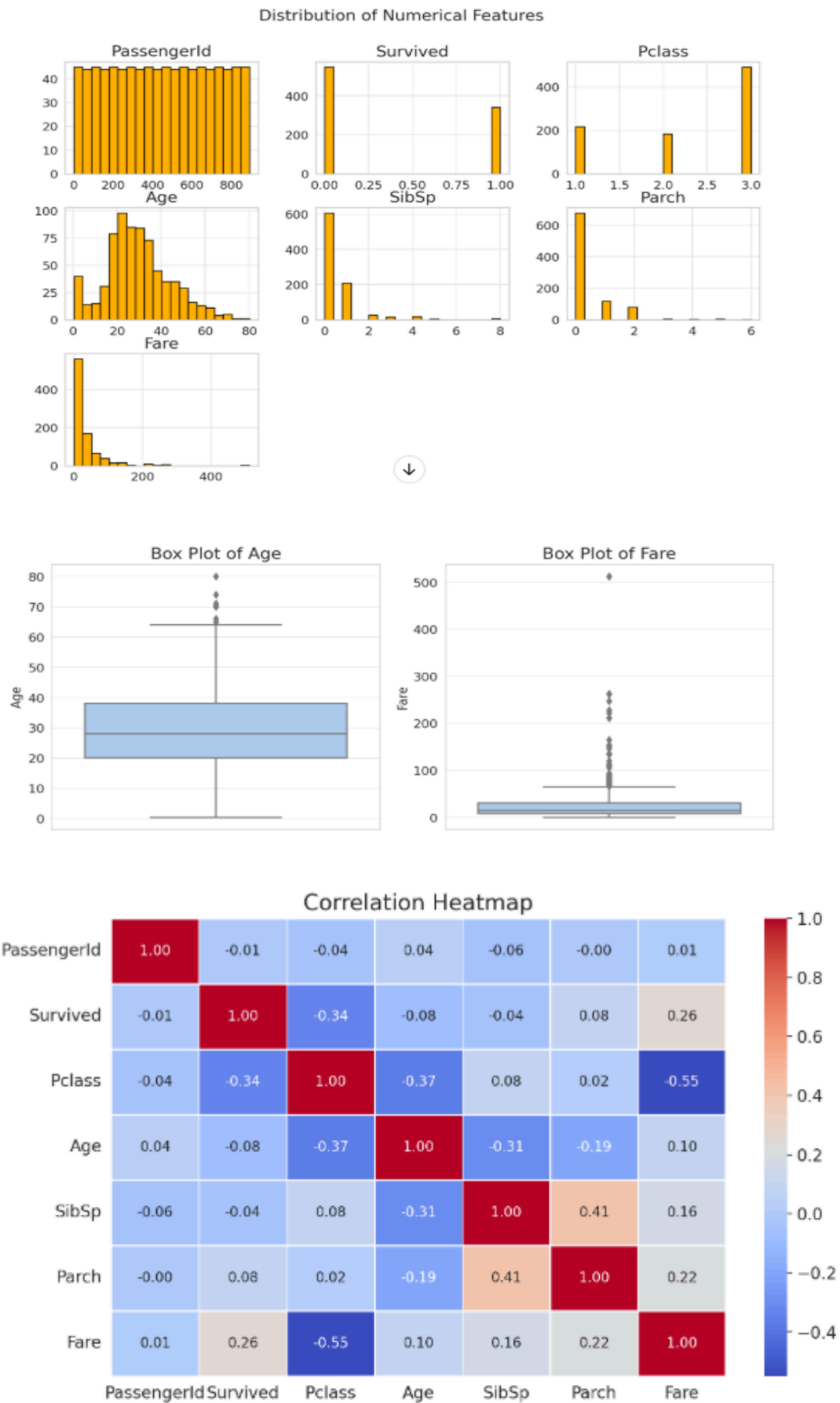
```
plt.imshow(wordcloud, interpolation="bilinear")
```

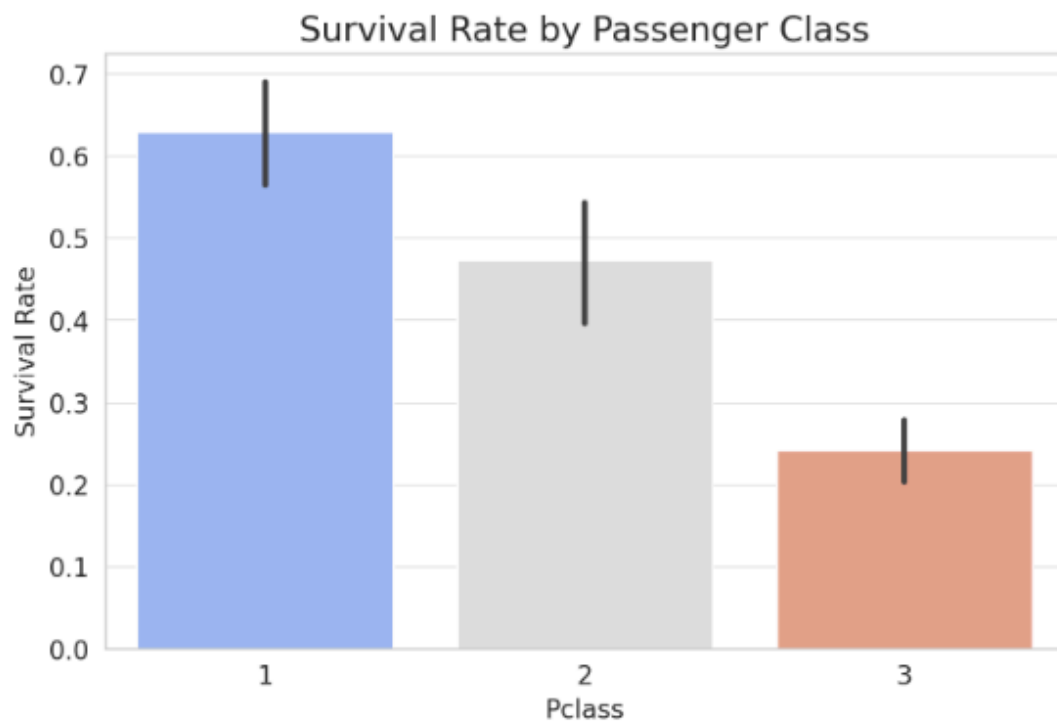
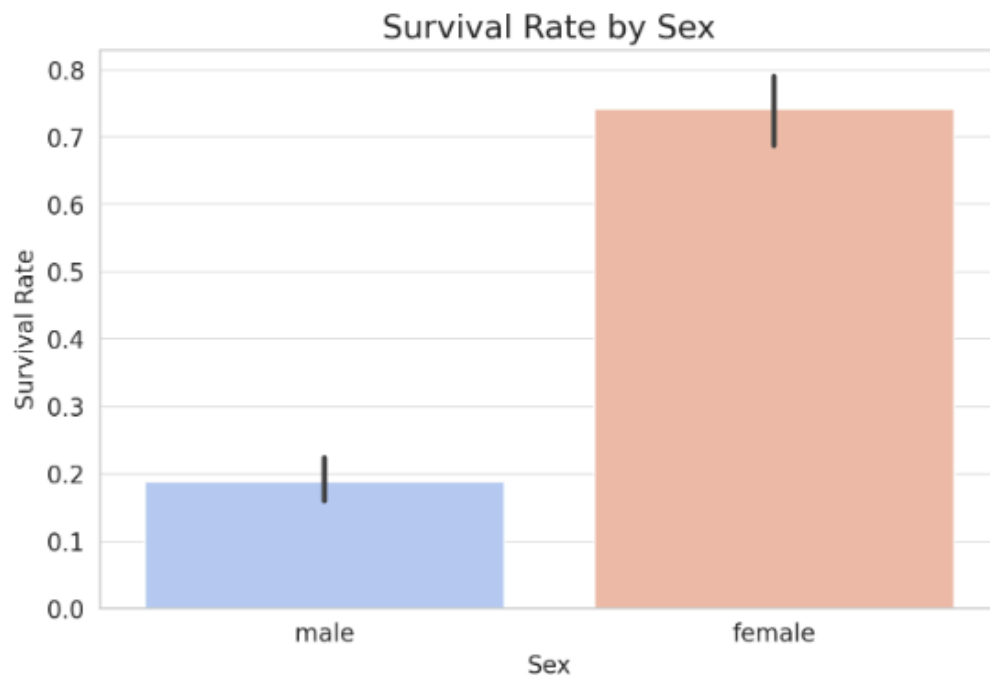
```
plt.axis("off")
```

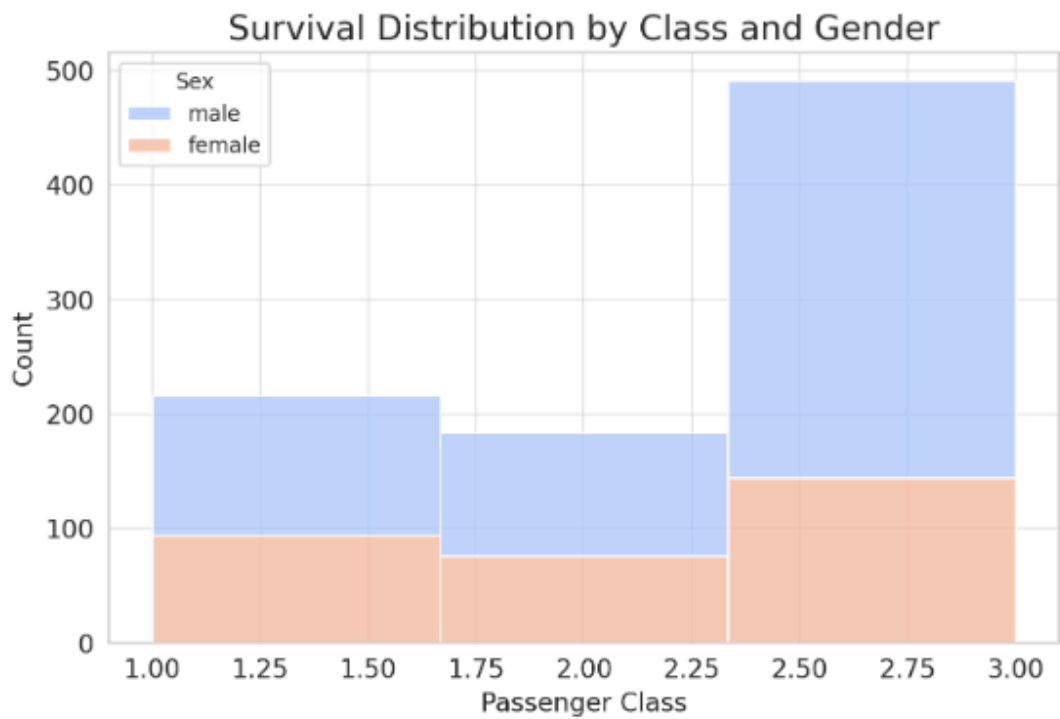
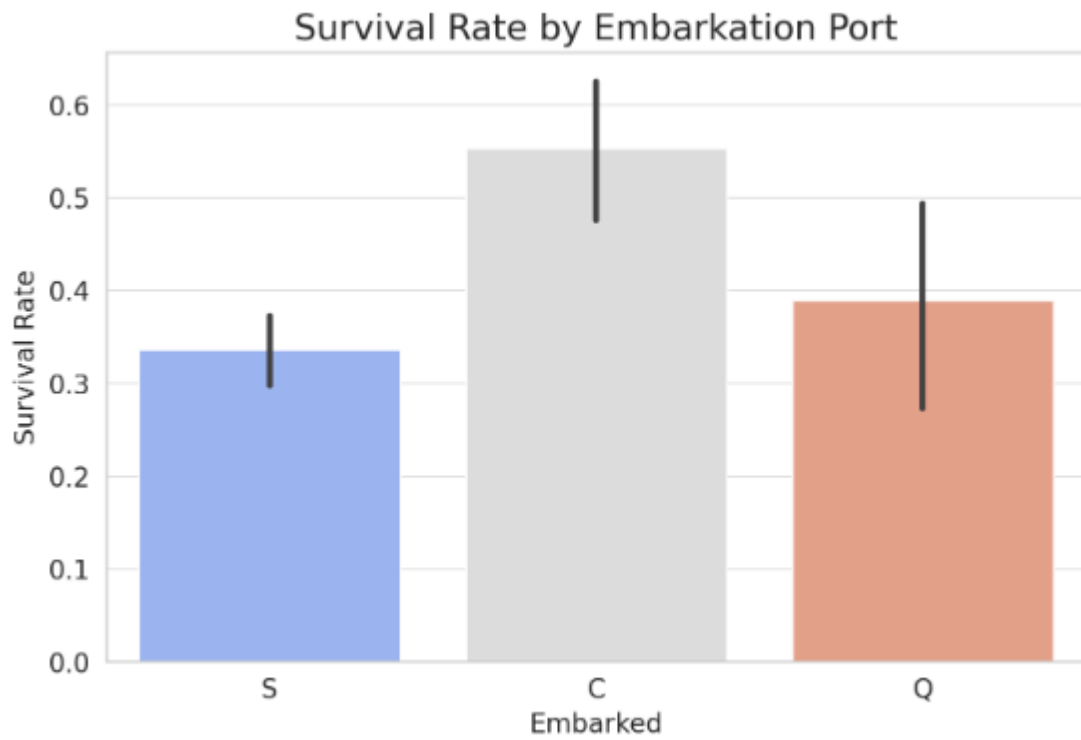
```
plt.title("Word Cloud of Passenger Names")
```

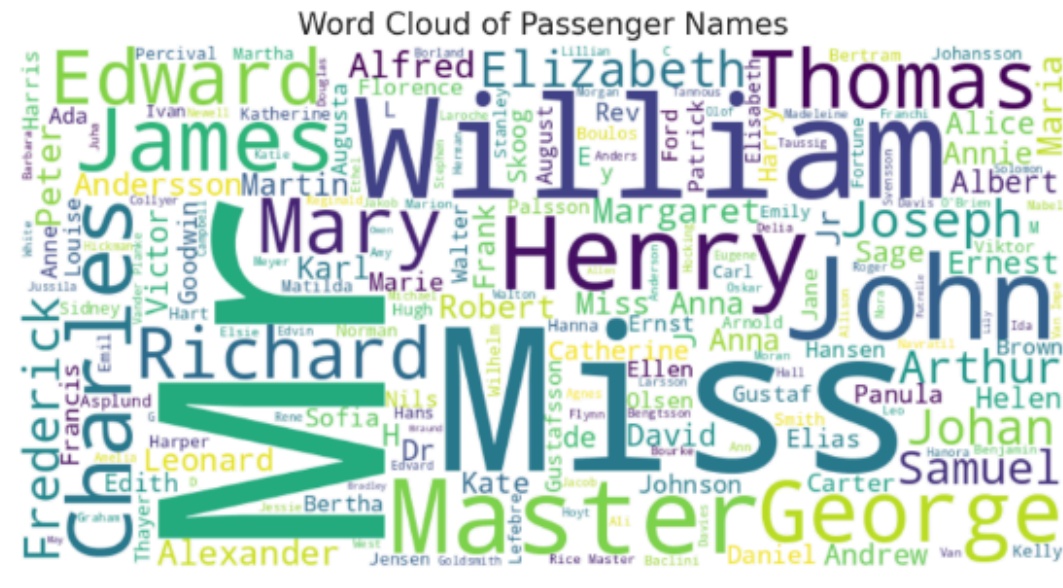
```
plt.show()
```

Output:









Conclusion:

- Women, first-class passengers, and those who embarked from Cherbourg had higher survival chances.
- Age alone wasn't a strong predictor of survival.
- Fare played a significant role, with higher-paying passengers more likely to survive.
- Most third-class passengers were male, contributing to the lower survival rate for men.
- Family groupings might have influenced survival rates, as seen in the word cloud.